

JPE Replication Report

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1 Paper Identification

ID	20211186
Author	Morales
Title	Firm Heterogeneity and the Impact of Immigration: Evidence from German Establishments
Iteration	2

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. There is a missing matlab toolbox from your readme.
2. Please improve info on where to put the ADO files.
3. You don't provide the code to generate figures like A5. You have to provide it.
4. We found a few other bugs in the code, please fix those.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☒ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☐ License for Code (Optional, but recommended)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.

☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
liab_lm_9314_v1_pers.dta (LIAB LM 9314)	no	yes	https://fdz.iab.de	yes	yes	yes
liab_lm_9314_v1_bhp_basis_v1.dta (LIAB LM 9314)	no	yes	https://fdz.iab.de	yes	yes	yes
iabbp_dta (IAB Establishment Panel, 2004–2012)	no	yes	https://fdz.iab.de	yes	yes	yes
SIEED_7518_v1.dta	no	yes	https://fdz.iab.de	yes	yes	yes
SIEED_7518_v1_bhp_basis_v1.dta	no	yes	https://fdz.iab.de	yes	yes	yes
cpigermany.dta (Destatis CPI)	yes	no	https://www-genesis.destatis.de/datenbank/online/statistic/61111/table/61111-0001	yes	yes	yes
local labor markets.dta (Kropp & Schwengler concordance)	yes	no	-	yes	yes	yes
Onet_Peri_Sparber_beruf_gr.dta	yes	no	https://www.aeaweb.org/articles?id=10.1257/app.1.3.135	yes	yes	yes
Validation_data.xlsx (aggregated IAB moments)	yes	no	-	yes	yes	yes
World Bank WDI series (SL.GDP.PCAP.EM.KD, SP.POP.1564.TO, NY.GDP.PCAP.PP.CD, SP.POP.TOTL)	yes	no	https://data.worldbank.org	yes	yes	yes
WIOT2012_October16_ROW.dta (WIOD 2012 release)	yes	no	https://www.rug.nl/ggdc/valuechain/wiod/wiod-2013-release	yes	yes	yes
OECD CSVs (wages, hours, GDP, dependent employees, Germany 2019)	yes	no	https://stats.oecd.org	yes	yes	yes

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☒ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

☐ No requirements specified

☒ Software Requirements specified as follows:

◦ 3.1 STATA

- Stata 16 or later. (liab_analysis/prog/master.do declares `version 16.`)
- User-written packages (install once from SSC):

```
ssc install reghdfe      // high-dimensional fixed-effects regression
ssc install ivreghdfe    // IV/2SLS version of reghdfe
ssc install ftools       // dependency of reghdfe/ivreghdfe
ssc install require      // package/version dependency checker
ssc install estout        // eststo / esttab / estout table export
ssc install grstyle      // graph styling (model_analysis plots only)
ssc install grc1leg2      // graph styling (model_analysis plots only)
```

ivreghdfe additionally relies on ivreg2 and ranktest; if `ssc install ivreghdfe` does not pull them in, install them as well: `ssc install ivreg2 ssc install ranktest`

The do-files in calibration_external/ use base Stata only (import Excel, merge, collapse) and require none of the above.

Also, make sure to add the ado files bartik_weight.ado, btsls.ado, and ch_weak.ado, which are not available via ssc install. The codes can be downloaded directly at <https://github.com/paulgp/bartik-weight/tree/master/code>

3.2 MATLAB

- MATLAB (R2019b or later recommended) with the Optimization Toolbox. The model solution and calibration use fsolve and fminsearch; the estimation helper files in liab_analysis/prog use fsolve and gamma().
- No additional MATLAB toolboxes are required.

☒ Computational Requirements specified as follows:

◦ 3.4 MEMORY AND STORAGE REQUIREMENTS

Restricted-data Stata analysis (liab_analysis, sieed_analysis): All computations on the confidential data are carried out on the servers of the FDZ. Replicators therefore do not need to provision memory or storage for this part of the analysis.

MATLAB model (model_analysis) and MATLAB helper files: The structural model is small and runs on any standard computer capable of running MATLAB itself. A machine meeting MathWorks' standard recommendation of 8 GB of RAM is more than sufficient. Model_analysis requires under 15 MB in total.

☒ Time Requirements specified as follows:

◦ 3.3 RUN TIME AND REPRODUCIBILITY

- Run time is data-dependent. The empirical part of the model takes close to 5 hours to run on the full data. On the genuine FDZ data, the bootstrap-based do-files (5,000 firm-clustered draws) and the model calibration (SMM via fminsearch) are the slow steps.
- Seeds are set (set seed 1234 in Stata; explicit bootstrap seeds were used).
- With the shipped TEST data the code runs to completion but outputs WILL NOT match the paper.

☐ Requirements are complete.

For missing requirements, see the list of required changes in [Section 9](#)

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

△ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 20
- Variables flagged in data: 243
- Code files with PII references: 85
- PII references in code: 1409

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	SIEED_7518_v1_bhp_basis_v1.dta	4	lat, son
Data	WIOT2012_October16_R0W.dta	1	country
Data	cpigermany.dta	1	lat
Data	iabbp_2004.dta	26	son, url
Data	iabbp_2005.dta	21	son, lat
Data	iabbp_2006.dta	21	son, url
Data	iabbp_2007.dta	19	son, lat
Data	iabbp_2008.dta	22	son, lat
Data	iabbp_2009.dta	20	son, lat
Data	iabbp_2010.dta	28	son, url, lat

File Type	File	Variables/References	PII Categories
Data	iabbp_2011.dta	26	son, url, lat
Data	iabbp_2012.dta	25	son, lat
Data	liab_lm_9314_v1_bhp_basis_v1.dta	3	lat, son
Data	liab_lm_9314_v1_pers.dta	4	son, gender, school, lat
Data	local labor markets.dta	13	name, loc
Data	oecd_weekly_hours_germany_2019_all_employees.csv	1	sex
Data	oecd_weekly_hours_germany_2019_full_time.csv	1	sex
Data	sieed_7518_v1.dta	1	son
Data	wdi_gdp_per_capita_ppp_2011.csv	3	country, name
Data	wdi_population_2011.csv	3	country, name

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-07-28T21:59:46.828

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any **windows** filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that **STATA** allows **/** to be a filepath separator on a windows platform, hence this is a requirement for *all* **STATA** applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
93	0	29	3	0

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum
/Brinatti_Morales_Replication_Package/model_analysis/code/model/homogeneous/Results_counterfactuals/.gitkeep	0.0	da39a3ee5
/Brinatti_Morales_Replication_Package/model_analysis/code/model/policies/Results_counterfactuals/.gitkeep	0.0	da39a3ee5
/Brinatti_Morales_Replication_Package/calibration_external/data/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/calibration_external/prog/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/liab_analysis/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/liab_analysis/data/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/liab_analysis/log/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/liab_analysis/orig/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/liab_analysis/prog/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/heterogeneous/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/heterogeneous/Equations_equilibrium/desktop.ini	0.0	3fa052fe4e

Filename	Size (MB)	Checksum
/Brinatti_Morales_Replication_Package/model_analysis/code/model/heterogeneous/Other_functions/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/heterogeneous/Results_counterfactuals/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/homogeneous/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/homogeneous/Equations_equilibrium/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/homogeneous/Other_functions/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/homogeneous/Results_counterfactuals/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/policies/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/policies/Equations_equilibrium/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/policies/Results_counterfactuals/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/stata_plots/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/data/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/data/inputs/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/data/raw/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/output/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/output/figures/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/heterogeneous/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/homogeneous/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/policies/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/output/tables/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/scripts/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/sieed_analysis/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/sieed_analysis/data/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/sieed_analysis/orig/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/sieed_analysis/prog/desktop.ini	0.0	3fa052fe4e
/Brinatti_Morales_Replication_Package/model_analysis/code/model/policies/Equations_equilibrium/utildet.m	0.0	5d7437846
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/homogeneous/Counterfactual_homogeneous_closed.mat	0.28	a4d4b42f6
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/homogeneous/Counterfactual_homogeneous_open.mat	0.29	832c177c39
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/heterogeneous/Counterfactual_heterogeneous_closed.mat	0.39	0f5482190f
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/policies/Counterfactual_to_compare_policies_top_onlyt.mat	0.4	3caa83c93
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/policies/Counterfactual_to_compare_policies_top10.mat	0.4	87b80aeaa
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/policies/Counterfactual_to_compare_policies_bottom_onlyt.mat	0.4	e62eed64d
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/heterogeneous/Counterfactual_heterogeneous_open.mat	0.41	c876f268a7
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/policies/Counterfactual_to_compare_policies_basef.mat	0.41	cbfaf0fb56
/Brinatti_Morales_Replication_Package/model_analysis/output/intermediate/policies/Counterfactual_to_compare_policies_nobasef.mat	0.42	2a521381f1

7.1.4 Zero Size Files Report

We found the following zero size files:

Filename	Size (MB)	Checksum (MD5)
/Brinatti_Morales_Replication_Package/model_analysis/code/model/heterogeneous/Results_counterfactuals/.gitkeep	0.0	da39a3ee5e6b4b0d3255bfef9560
/Brinatti_Morales_Replication_Package/model_analysis/code/model/homogeneous/Results_counterfactuals/.gitkeep	0.0	da39a3ee5e6b4b0d3255bfef9560
/Brinatti_Morales_Replication_Package/model_analysis/code/model/policies/Results_counterfactuals/.gitkeep	0.0	da39a3ee5e6b4b0d3255bfef9560

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/Brinatti_Morales_Replication_Package/sieed_analysis/orig/sieed_7518_v1.dta	149.31
/Brinatti_Morales_Replication_Package/liab_analysis/orig/liab_lm_9314_v1_pers.dta	867.08

7.2 Code description

There are 34 **.do** files. 1 master file at the root, 1 file in the calibration_external folder, 20 for the liab_analysis, 7 for the sieed_analysis, 1 for the model_analysis, and 4 for the stat_plots.

There are 50 **.m** files. 1 master file in the model_analysis, 2 files in the liab_analysis, 1 to export the latex tables, 19 for the heterogeneous-firm model, 15 for the homogeneous (within-sector) model, and 12 for the policy counterfactuals.

The below table presents programs which produce figures or tables.

Exhibit	Folder	Program
Table 1	liab_analysis	19_moments_calculation.do
Table 1 (model side) / Table 2	model_analysis	code/model/Export_latex_tables.m
Tables A1, A2	liab_analysis	17_empirical_facts.do
Tables A3–A5	sieed_analysis	2a_results_event_studies.do
Tables A6–A8	liab_analysis	18_immigrant_comparative_advantage.do
Table C1	liab_analysis	10_epsilon_estimation.do
Tables C2, C4	liab_analysis	15_epsilon_shift_share.do
Table C3	liab_analysis	11_epsilon_pre_trend_test.do
Table C5	liab_analysis	12_epsilon_firm_characteristics.do
Table C6	liab_analysis	16_estimate_nu.do (+ MATLAB nu files)
Table C7	liab_analysis / model_analysis	19_moments_calculation.do / Export_latex_tables.m
Tables C8–C13	liab_analysis	8_validation_regressions.do
Tables D1–D3	model_analysis	code/model/Export_latex_tables.m
Kappa (Appendix C.1)	liab_analysis	9_kappa_estimation.do
Figure 1	liab_analysis	17_empirical_facts.do
Figure 2	sieed_analysis	2a_results_event_studies.do
Figure 3	liab_analysis / model_analysis	8_validation_regressions.do / master.do (Validation_cross_CI_heterog.do)
Figure 4	model_analysis	master.do (Counterfactual_plots_heterog.do)
Figures A1–A4, A6	liab_analysis	17_empirical_facts.do
Figure A5	sieed_analysis	2c_results_figA5.do

Exhibit	Folder	Program
Figure A7	sieed_analysis	2b_results_figA7.do
Figure A8	sieed_analysis	2a_results_event_studies.do
Figure A9	liab_analysis	18_immigrant_comparative_advantage.do
Figure C1	liab_analysis	14_epsilon_sigma_histogram.do
Figures D1, D2	model_analysis	master.do (Counterfactual_plots_policies.do)
Calibration moment GDP_RoW_Ger (~0.32)	calibration_external	gdp_per_worker_*.do
Calibration moment tradeable share (~0.68)	calibration_external	tradeable_consumption_share.do

cloc		github.com/AIDanial/cloc v 2.06 T=0.07 s (1431.3 files/s, 233364.0 lines/s)		
Language	files	blank	comment	code
Stata	34	2920	608	6509
MATLAB	49	1095	1246	2093
Text	3	205	0	1499
CSV	7	0	0	547
TeX	9	0	9	117
Markdown	2	29	0	79
---	---	---	---	---
SUM:	104	4249	1863	10844

☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Macbook Air 2020
- Apple M1 Chip
- 8 GB RAM
- Mac OS Tahoe 26.5.2
- StataMP 19.5
- Matlab R2026a

8 Replication

Overall the script runs smoothly with 1 issue.

When running the file, an error appeared related to permissions:

```
. log using "${log}/master.log", replace
(file
  /Users/mac/JPE-replications/JPE-Morales-20211186/replication-package/Brinatti_Morales_Replic
  > ation_Package/liab_analysis/log/master.log not found)
file
  /Users/mac/JPE-replications/JPE-Morales-20211186/replication-package/Brinatti_Morales_Replic
  > ation_Package/liab_analysis/log/master.log could not be opened
r(603);
```

Solution:

I ran a background command (chmod -R u+w) on the entire Brinatti_Morales_Replication_Package folder to grant write permissions for all files inside it.

After fixing, another error appeared:

mvnrnd requires Statistics and Machine Learning Toolbox.

Error in Heterogeneity (line 56)

```
normal_draw = exp(mvnrnd(mu, var, N_g));
```

```
~~~~~
Error in Counterfactual_heterogeneous_open (line 14)
SS = Heterogeneity(PP);
```

```
~~~~~
```

Error in run (line 100)

```
evalin('caller', strcat(scriptStem, '));
```

```
~~~~~
```

Error in master>run_isolated (line 100)

```
run(script_name);
```

```
~~~~~
```

Error in master (line 33)

```
run_isolated('Counterfactual_heterogeneous_open.m');
```

```
~~~~~
```

Error in run (line 100)

```
evalin('caller', strcat(scriptStem, '));
```

```
~~~~~
```

```
}
```

After installing Machine Learning Toolbox in Matlab, the code ran to completion.

9 Findings

9.1 Missing Requirements

The provided test data took approximately 1 hour to run on the replicator's machine (Stata analysis: ~48 minutes, MATLAB model: ~8 minutes). Please add this information to the README.

[REQUIRED] Please amend README to contain complete requirements.

9.2 Data Preparation Code

The master file `master_file.do` ran without issues after installing the Statistics and Machine Learning Toolbox in Matlab.

9.3 Tables and Figures

Exhibit	Folder	Program	Reproduced?	Comment
Table 1	liab_analysis	19_moments_calculation.do	yes	rounding discrepancy on the which equals 30.82 in the paper and 30.83 in the reproduced file.
Table 2	model_analysis	code/model/Export_latex_tables.m	yes	-
Tables A1, A2	liab_analysis	17_empirical_facts.do	yes	differences appear because of confidentiality of the data
Tables A3–A5	sieed_analysis	2a_results_event_studies.do	yes	differences appear because of confidentiality of the data
Tables A6–A8	liab_analysis	18_immigrant_comparative_advantage.do	yes	differences appear because of confidentiality of the data
Table C1	liab_analysis	10_epsilon_estimation.do	yes	differences appear because of confidentiality of the data
Tables C2, C4	liab_analysis	15_epsilon_shift_share.do	yes	differences appear because of confidentiality of the data
Table C3	liab_analysis	11_epsilon_pre_trend_test.do	yes	differences appear because of confidentiality of the data
Table C5	liab_analysis	12_epsilon_firm_characteristics.do	yes	differences appear because of confidentiality of the data

Exhibit	Folder	Program	Reproduced?	Comment
Table C6	liab_analysis	16_estimate_nu.do (+ MATLAB nu files)	yes	differences appear because of confidentiality of the data
Table C7	liab_analysis / model_analysis	19_moments_calculation.do / Export_latex_tables.m	yes	rounding discrepancies (paper vs reproduced file). , T: simulated 1.31 vs 1.32 , NT: simulated 1.43 vs 1.42 data 1.58 vs 1.59. , T: simulated 0.027 vs 0.026. Share of firms hiring immigrants, T: simulated 0.62 vs 0.61.
Tables C8–C13	liab_analysis	8_validation_regressions.do	yes	differences appear because of confidentiality of the data
Tables D1–D3	model_analysis	code/model/Export_latex_tables.m	yes	-
Kappa (Appendix C.1)	liab_analysis	9_kappa_estimation.do	yes	differences appear because of confidentiality of the data
Figure 1	liab_analysis	17_empirical_facts.do	no	No image files produced by code; only underlying data tables.
Figure 2	sieed_analysis	2a_results_event_studies.do	no	No image files produced by code; only underlying data tables.
Figure 3	liab_analysis / model_analysis	8_validation_regressions.do / master.do (Validation_cross_CI_heterog.do)	yes	differences appear because of confidentiality of the data
Figure 4	model_analysis	master.do (Counterfactual_plots_heterog.do)	yes	-
Figures A1–A4, A6	liab_analysis	17_empirical_facts.do	no	No image files produced by code; only underlying data tables.
Figure A5	sieed_analysis	2c_results_figA5.do	no	No image files produced by code; only underlying data tables.
Figure A7	sieed_analysis	2b_results_figA7.do	no	No image files produced by code; only underlying data tables.
Figure A8	sieed_analysis	2a_results_event_studies.do	no	No image files produced by code; only underlying data tables.
Figure A9	liab_analysis	18_immigrant_comparative_advantage.do	no	No image files produced by code; only underlying data tables.
Figure C1	liab_analysis	14_epsilon_sigma_histogram.do	yes	differences appear because of confidentiality of the data
Figures D1, D2	model_analysis	master.do (Counterfactual_plots_policies.do)	yes	-
Calibration moment GDP_RoW_Ger (~0.32)	calibration_external	gdp_per_worker_*.do	yes	differences appear because of confidentiality of the data
Calibration moment tradeable share (~0.68)	calibration_external	tradeable_consumption_share.do	yes	differences appear because of confidentiality of the data

9.4 In-Text Numbers

- ☐ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code
- ☒ The below in-text numbers have been verified.

Section 6.1 aggregates and dollar gains

"that the welfare of native workers would increase by 0.11%, representing \$1.8 billion for the aggregate economy."

(...)

"The welfare gains of firm owners (1.31%) are larger than those of native workers"

(...)

"Total income, which includes earnings for all workers and profits, increases by 1.28% or \$22.5 billion."

Section 6.3 decomposition

"which is 9% (0.010 pp) larger than"

(...)

"implying that terms-of-trade effects reduce the gains from immigration by 0.004 pp (approximately 39% of the total dampening effect)."

(...)

"The remaining 61% of the total dampening effects (0.006 pp) is due to the heterogeneity in immigrant shares across firms within sectors, as illustrated in equation 16."

9.5 Other Issues

- In Instructions to Replicators, please add the info that if Matlab is added to PATH, everything can be replicated using master_file.do
- There's no info on where the .ado files should be located
- The package should be unzipped directly to **replication-package** folder
- There is no paper in the paper-appendices folder, please relocate
- Please provide the file in open format: **model_analysis/data/inputs/Validation_data.xlsx**
- There's an unnecessary tabulate in readme in table line 459 (look at the screenshot)

459	EXHIBIT	FOLDER	PROGRAM	CONFIDENTIAL DATA?
460	-----	-----	-----	-----
461	Table 1 (data side)	liab_analysis	19_moments_calculation.do	Yes
462	Table 1 (model side) / Table 2	model_analysis	code/model/Export_latex_tables.m	No
463	Table A1, A2	liab_analysis	17_empirical_facts.do	Yes

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☒ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
- ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.

- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☒ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

10.2 Potential Personal Identifiable Information (PII)

△ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this – please check!

Summary:

- Data files with PII indicators: 20
- Variables flagged in data: 243
- Code files with PII references: 85
- PII references in code: 1409

10.2.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	SIEED_7518_v1_bhp_basis_v1.dta	4	lat, son
Data	WIOT2012_October16_ROW.dta	1	country
Data	cpigermany.dta	1	lat
Data	iabbp_2004.dta	26	son, url
Data	iabbp_2005.dta	21	son, lat
Data	iabbp_2006.dta	21	son, url
Data	iabbp_2007.dta	19	son, lat
Data	iabbp_2008.dta	22	son, lat
Data	iabbp_2009.dta	20	son, lat
Data	iabbp_2010.dta	28	son, url, lat
Data	iabbp_2011.dta	26	son, url, lat
Data	iabbp_2012.dta	25	son, lat
Data	liab_lm_9314_v1_bhp_basis_v1.dta	3	lat, son
Data	liab_lm_9314_v1_pers.dta	4	son, gender, school, lat
Data	local_labor_markets.dta	13	name, loc
Data	oecd_weekly_hours_germany_2019_all_employees.csv	1	sex
Data	oecd_weekly_hours_germany_2019_full_time.csv	1	sex
Data	sieded_7518_v1.dta	1	son
Data	wdi_gdp_per_capita_ppp_2011.csv	3	country, name
Data	wdi_population_2011.csv	3	country, name

See [Appendix](#) for detailed listing of all flagged instances.